

PART 1: 1.5 points each (30 points total)

1. C
2. B
3. B
4. Any of the following 2 are acceptable
 - Increases turbidity
 - Smothers benthic organisms
 - Increases temperature
 - Transports attached pollutants/nutrients
 - Reduces light penetration
 - Clogs fish gills
 - Disrupts feeding and reproduction
5. B
6. B
7. Wastewater (or Sewage)
8. B
9. Potable is for drinking/removing pathogens; Wastewater is for discharging/reducing environmental pollutants
10. C
11. A
12. A
13. D
14. Coagulation is the rapid mixing of chemicals (like alum) to neutralize charges and start clump formation.

Flocculation is the slow, gentle mixing that encourages those small clumps to collide and grow into heavy "floc" that can settle out.

15. C
16. B
17. C
18. B
19. D
20. A

PART 2 (3 points each) (30 points total):

21. B
22. C
23. A: Class 1 | B: Class 2 | C: Class 5 (or 4) | D: Class 1
24. A
25. They show cumulative water quality impacts over time because they are sessile.
26. D
27. B
28. A

29. Accept anyone of the following

- Filter feeding
- Shredding
- Scraping/Grazing
- Predation

30. C

Part III: Water Monitoring and Analysis (3 points each) (30 points total)

31. B

32. B

33. B

34. They indicate fecal contamination and the potential presence of pathogens.

35. C

36. B

37. High turbidity blocks sunlight, reducing plant photosynthesis/growth.

38. B

39. C

40. It reveals trends and fluctuations that a single snapshot misses. Revealed seasonal variation and reduces influence of abnormality.

Part IV: Salinometer Testing

- If device is present and functional : 5 points
- Measurement: Value within 1 percent of real (5 pts)

Real value: 6% salinity